

PLAYING TOUGHER FOR ONE’S SIZE: CALIBRATED SIZE ABOVE EXPECTED IN THE NHL

RENTO SAIJO¹, SIAN LEE², JEFF MOHER¹, YEONGJUN SEO^{3*}

¹ Connecticut College; ² Michigan State University; ³ University of Tennessee, Knoxville

* Corresponding author: yseo4@utk.edu

Introduction

In hockey, physical play involves contact and battles for the puck and space, but body size and hit counts alone do not quantify whether a player’s physical play exceeds or falls below expectations for their size. We develop calibrated size above expected (CSAx), a position-specific measure of physical play relative to listed size, to operationalize *playing tougher for one’s size*. We examine its agreement with scouting reports and links to special-teams deployment, postseason hitting, and next-season continuation.

Methods

NHL event data and All Three Zones’ manually tracked five-on-five data were combined, covering 1,485 forward- and 829 defenseman-seasons from 2021–22 through 2024–25. Separate ridge regressions for forwards and defensemen estimate behavior-implied size by predicting standardized size derived from listed height and weight. Predictors include six shared direct measures (e.g., hits delivered and received), six forward indirect measures (e.g., dump-in recoveries), and three defenseman indirect measures (e.g., entry-denial share). Held-out predictions are calibrated against listed size, and residuals are standardized within position and season:

$$\text{CSAx} = \frac{r - \bar{r}}{s_r}, \quad r = \hat{S} - \hat{m}(S).$$

Here, S is standardized listed size, $\hat{S}$ is behavior-implied size, $\hat{m}(S)$ is the expected behavior-implied size at S , r is the calibrated residual, and $\bar{r}$ and s_r are the mean and standard deviation of these residuals within each position and season. Higher CSAx indicates greater behavior-implied size after accounting for listed size. One rater, blinded to identity and CSAx, codes mentions of physical traits (e.g., checking, battles) in official NHL draft-year scouting reports published before the studied seasons. Linear models assess special-teams shares and returner ice time; logistic models assess next-season continuation (at least 300 NHL minutes); and Poisson models compare hit rates within player-seasons with ice-time offsets. Analyses adjust for listed size, age and age squared, current-season games dressed and ice time per game, five-on-five points per 60, relative shot attempts, and season. We compare regular-season games excluded from CSAx construction with each team’s first four postseason games, a common window regardless of advancement.

Results

Players with physical-trait mentions in earlier scouting reports have higher mean CSAx than those without, by 0.87 for forwards and 0.65 for defensemen (Figure 1).

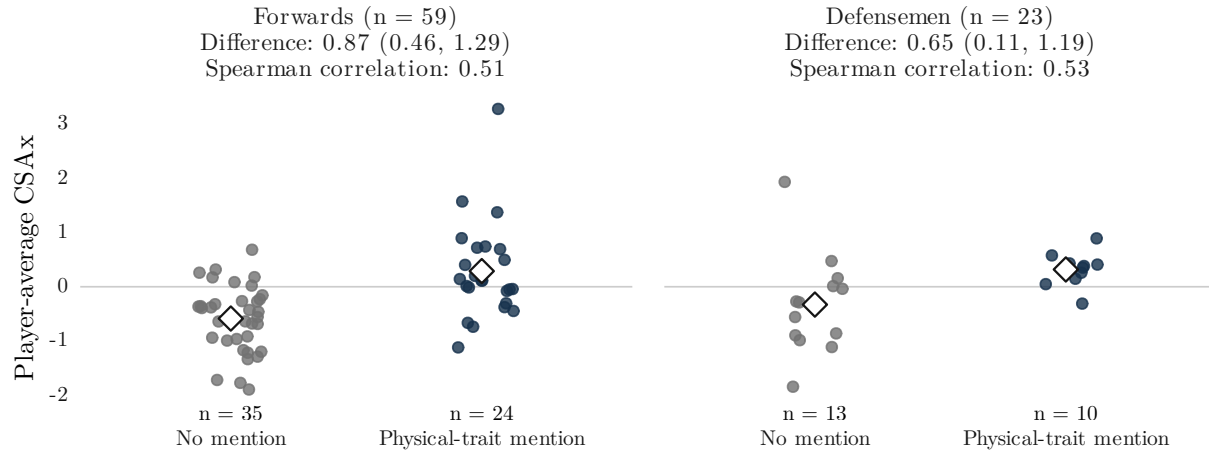


Figure 1. Player-average CSAx by whether an earlier scouting report mentions active physical traits. Points represent players; diamonds mark group means. Mean differences, with 95% confidence intervals, suggest general agreement with earlier scouting descriptions.

Higher CSAx is associated with a smaller share of players' total ice time on power plays and a larger share killing penalties in both positions (Table 1). Among forwards, higher CSAx is associated with a lower postseason-to-regular-season hit-rate ratio; the defensemen estimate is uncertain. Higher CSAx also predicts next-season continuation, but not ice time among returners.

Table 1. Adjusted comparisons of CSAx +1 versus -1 (95% player-clustered confidence intervals in parentheses). Postseason estimates are ratios of postseason-to-baseline hit-rate ratios (+1 divided by -1); all other estimates are differences (+1 minus -1).

Outcome	Forwards	Defensemen
Power-play share (percentage points)	-1.22 (-1.78, -0.67)	-0.88 (-1.51, -0.25)
Penalty-kill share (percentage points)	1.40 (0.74, 2.06)	1.31 (0.74, 1.89)
Postseason hit-rate (ratio)	0.78 (0.69, 0.88)	0.90 (0.77, 1.05)
Next-season continuation (percentage points)	2.92 (0.03, 5.81)	4.23 (0.60, 7.86)
Ice time (min/game)	-0.18 (-0.36, 0.00)	-0.14 (-0.43, 0.14)

Conclusion

CSAx quantifies physical play relative to listed size, measuring *playing tougher for one's size*. The high-CSAx defenseman without a physical-trait mention highlights a scouting application: identifying smaller players whose engagement may be overlooked by absolute size. Agreement with scouting and associations with penalty-kill use and next-season continuation support relevance to specialized roles. Continuation without greater ice time among returners suggests CSAx captures playing style relevant to NHL participation. Alongside hits and scoring, CSAx adds a size-relative perspective on toughness and roster fit.